

WIPI2 (Q9Y4P8) in Pexophagy, Nucleophagy and Glycophagy — Focused Curation Report

Gene: WIPI2 (human, NCBITaxon:9606) · **UniProt:** Q9Y4P8 · **PANTHER family:** PTHR11227 (WIPI/Atg18 PROPPIN family), subfamily SF27 **Focus:** function_assignment · **Hypothesis slug:** selective-pexophagy-nucleophagy-and-glycophagy **Terms under review:** GO:0000425 pexophagy · GO:0044804 nucleophagy · GO:0061723 glycophagy (all currently IBA)

Executive Judgment

Verdict: Partially supported — retain as non-core, shared-machinery participation; do not treat as WIPI2-specific selective functions.

The seed hypothesis is essentially correct on mechanism. Human WIPI2's only experimentally supported activity is a **generic core step of autophagosome biogenesis**: it senses PI3P and recruits the ATG12–ATG5–ATG16L1 (E3) complex, allosterically activating LC3/GABARAP lipidation

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Pexophagy, glycophagy (both defined by GO as "by macroautophagy") and mammalian macro-nucleophagy all build canonical LC3/GABARAP double-membrane autophagosomes and therefore **mechanistically require this WIPI2 step**. In that sense WIPI2 "participates" — exactly as it demonstrably does in Salmonella xenophagy, where WIPI2b coats the bacterial membrane and recruits ATG12-5-16L1

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However, three important caveats keep this at **partially supported** / **non-core** rather than "supported":

1. **No target-specific experiment exists for human WIPI2** in any of the three pathways. All three annotations are phylogenetic inferences (**IBA**, **ECO:0000318**, **GO_Central**, **GO_REF:0000033**), not IDA/IMP on WIPI2.
2. **WIPI2 is not the cargo/selectivity receptor** for any of the three: pexophagy uses NBR1/p62 + ubiquitinated PEX5 (+PEX2 E3); mammalian nucleophagy uses a direct LC3–LMNB1 interaction; glycophagy uses STBD1 (CBM20+LIR→GABARAPL1/RB1CC1). The seed's own framing — "may contribute common phagophore/ATG16L1 adaptor machinery without being the selective cargo receptor" — is confirmed.
3. **The IBA seeds are loss-of-function "general requirement" experiments** on the yeast/fly ortholog acting as core machinery, and the same blanket annotation was propagated identically to all four human paralogs (WIPI1/2/3/4), which have functionally diverged (ATG16L1 arm vs ATG2 arm). IBA cannot resolve this.

Bottom line for the curator: The activity is real but generic. These BP terms are defensible as "required-for/part-of" shared machinery, but they are **not** evidence that WIPI2 has a peroxisome-, nucleus-, or glycogen-specific function. If the review's convention is to record only pathway-specific or core functions, these three should be flagged **non-core** / **generalizable to macroautophagy (GO:0016236)** / **autophagosome assembly (GO:0000045)**, which WIPI2 *does* hold by direct experiment.

Evidence Matrix

#	Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
1	QuickGO/ GO_Central (GO_REF:0000033)	review/ database	Qualifies	Are WIPI2's pexophagy/ nucleophagy/ glycophagy annotations experimental?	All three (+mitophagy) are IBA only ; no IDA/IMP/EXP on Q9Y4P8	Human WIPI2 annotation set (50 annotations)
2	24954904 (Dooley 2014)	direct assay + interaction	Supports (shared machinery)	WIPI2 molecular function	ATG16L1 directly binds WIPI2b; WIPI2b is PI3P effector required for LC3 conjugation; also required for autophagic clearance of <i>Salmonella</i>	Human cell MEFs, GUV
3	32437499 (Fracchiolla 2020)	direct assay (reconstitution)	Supports (shared machinery)	WIPI2 activates LC3 lipidation	On GUVs, LC3 lipidation strictly requires PI3P-dependent WIPI2, which allosterically activates the E3 complex	In vitro reconstitution
4	20505359 (Polson 2010)	localization + mutant	Supports (core function)	WIPI2 localizes/ regulates lipidation	WIPI2 at omegasome- anchored phagophores; positively regulates LC3 lipidation (IDA/ IMP source for CC/BP)	Human cell

#	Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
5	11739783 (yeast ATG18)	mutant phenotype (IMP)	Qualifies (IBA seed, pexophagy)	Is the Atg18 ortholog required for pexophagy?	<i>S. cerevisiae</i> Atg18 required for pexophagy	Yeast
6	18701704; 22768199 (yeast ATG18/ATG21)	mutant phenotype (IMP)	Qualifies (IBA seed, nucleophagy)	Atg18/Atg21 required for nucleophagy (PMN)	Core Atg cohort incl. Atg18 required for piecemeal microautophagy of the nucleus	Yeast
7	24265594 (Zirin/Perrimon, <i>Drosophila</i>)	mutant phenotype (IMP)	Qualifies (IBA seed, glycophagy)	Is Atg18a required for glycogen breakdown?	Whole core Atg set (incl. Atg18a) required for autophagic glycogen degradation	<i>Drosophila</i>
8	26524528; 26654219 (Dou 2015/2016)	direct assay + mutant	Competing (nucleophagy mechanism)	What confers nuclear selectivity?	Direct LC3–LMNB1 interaction targets lamin-associated domains; drives oncogene-induced senescence	Human primary cel
9	39236246 (Zhang 2024)	structural + direct assay	Competing (glycophagy receptor)	What confers glycogen selectivity?	STBD1 CBM20 binds glycogen; LIR binds GABARAPL1	Human, structures

#	Citation (PMID)	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context
					and recruits RB1CC1/FIP200	
10	36541703; 37493040; 39765694	mutant/ localization	Competing (pexophagy receptor)	What confers peroxisome selectivity?	NBR1/p62 + ubiquitinated PEX5, PEX2 E3 drive mammalian pexophagy	Human/ zebrafish/ra
11	QuickGO paralog scan	computational/ database	Qualifies (paralog over- propagation)	Is the IBA paralog- specific?	Identical IBA (pexo/nucleo/ glyco/mito) on WIPI1, WIPI2, WIPI3/WDR45B, WIPI4/WDR45	Human paralogs

GO Curation Implications (leads — require curator verification)

GO term	Aspect	Current	Evidence for WIPI2	Lead recommendation
GO:0000425 pexophagy	BP	IBA	Ortholog general-requirement (yeast Atg18); no human WIPI2 experiment; not the receptor	Retain as non-core OR generalize. Legitimate IBA, but flag as shared-machinery/"required-for" rather than a WIPI2-specific function. If the review disallows unverified selective-autophagy terms, generalize to GO:0016236/GO:0000045.
GO:0044804 nucleophagy	BP	IBA	Ortholog requirement in yeast PMN; mammalian selectivity is LC3–LMNB1, not WIPI2	Retain as non-core. Definition is broad (not restricted to yeast PMN), so IBA is not term-inappropriate, but treat as shared machinery.
GO:0061723 glycophagy	BP	IBA	Within-family experimental seed = Drosophila Atg18a (P 24265594); mammalian receptor is STBD1	Retain as non-core. IBA is phylogenetically valid; not WIPI2-cargo-specific.
GO:0016236 macroautophagy / GO:0000045 autophagosome assembly	BP	IMP/IDA (direct)	Direct experiments (P 20505359 P 24954904 P 28561066)	Retain as core — this is WIPI2's primary, directly supported BP.
GO:0032266 PI3P binding;	MF	IDA/IBA		

GO term	Aspect	Current	Evidence for WIPI2	Lead recommendation
GO:0030674 adaptor activity			Direct (P 20505359 P 28561066) 	Retain as core MF — more informative than "protein binding".

Recommended framing: WIPI2's *core* annotations are the PI3P-binding/ATG16L1-adaptor MF and macroautophagy/autophagosome-assembly BP. The three selective-autophagy BP terms are **downstream, pathway-context consequences of that same single molecular step**, delivered via IBA; they should be labeled **non-core / shared-machinery**, not removed as wrong (each has a valid within-family experimental seed), and certainly not read as cargo-recognition functions.

Mechanistic Scope

- **Immediate molecular function (direct):** PI3P-dependent β -propeller effector that recruits/allosterically activates the ATG12–ATG5–ATG16L1 E3 to lipidate LC3/GABARAP onto the phagophore
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Location: phagophore assembly site / omegasome / autophagosome membrane (IDA).
- **Downstream / pathway-context (inferred):** Because pexophagy and glycophagy are *macroautophagy* (GO definitions explicitly state "by macroautophagy") and mammalian nucleophagy of lamin B1 is autophagosome-mediated, loss of WIPI2 would impair all of them — but this is a **general requirement**, identical to losing any core Atg factor, not a peroxisome/nucleus/glycogen-specific activity.
- **Not attributable to WIPI2:** cargo selection (NBR1/p62/ubiquitin-PEX5; LC3–LMNB1; STBD1–glycogen). These provide the selectivity; WIPI2 provides the shared build step.

Conflicts and Alternatives

- **Paralog over-annotation:** All four human WIPI paralogs carry identical IBA for the four selective terms. WIPI1/WIPI2 operate in the ATG16L1-recruiting arm; WIPI3(WDR45B)/WIPI4(WDR45) operate with ATG2 in lipid transfer. Blanket IBA cannot distinguish these — a database carry-over artifact, though not necessarily biologically false (all are required for autophagosome formation).
 - **Organism-specific seeds:** The pexophagy/nucleophagy seeds are yeast (piecemeal microautophagy for PMN); glycophagy seed is *Drosophila*. Extrapolation to human WIPI2 is inference, not observation.
 - **Competing selectivity mechanisms:** For each pathway, the literature attributes selectivity to dedicated receptors, not WIPI2 — so any curation implying WIPI2 confers selective recognition would be an over-statement.
 - **"General requirement vs direct shared step":** The seed asks to distinguish these. The available evidence is loss-of-function *requirement* (IMP) plus a mechanistic *direct step* (PI3P→ATG16L1→LC3) that is generic; there is no experiment isolating a WIPI2 step *within* pexophagy/nucleophagy/glycophagy specifically.
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Knowledge Gaps

1. **No human WIPI2 loss-of-function in any of the three pathways.** Checked QuickGO (0 experimental annotations). Matters because IBA alone is being asked to justify three BP terms. Resolve with WIPI2 KO/KD + peroxisome (PMP70/PEX14), lamin B1, or glycogen turnover readouts.
2. **Paralog specificity.** Checked annotation blanketing across WIPI1/2/3/4. Matters because curators may want paralog-resolved terms. Resolve with paralog-specific KO rescue in each selective assay.
3. **Whether nucleophagy of lamin B1 requires WIPI2 at all** vs. proceeding through a non-canonical/WIPI-independent route. Checked: Dou papers implicate LC3 directly but did not test WIPI2. Resolve with WIPI2 dependence test in RAS-induced senescence lamin B1 degradation.
4. **Glycophagy dependence on WIPI2 specifically in mammals** (*Drosophila* seed only). Resolve with STBD1-based glycophagy flux assay under WIPI2 depletion.

Discriminating Tests

1. **WIPI2 depletion + selective-flux readouts:** pexophagy (peroxisome number/PMP70, HaloTag-PTS1 processing), nucleophagy (lamin B1 degradation in oncogene-induced senescence,

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model), glycopagy (glycogen/PAS or 2-NBDG flux, STBD1 reporter). Expectation if "shared machinery only": all three blocked to the same degree as bulk autophagy, with no additional selectivity phenotype.

2. **Separation-of-function ATG16L1-binding mutant of WIPI2** (WIPI2b residues that bind ATG16L1 vs PI3P): should block all three pathways coordinately, confirming the single shared step.
3. **Paralog swap/rescue:** test whether WIPI3/WIPI4 (ATG2 arm) rescue each selective pathway in WIPI2-null cells, to resolve paralog-specific claims.
4. **Proximity/interaction proteomics of WIPI2 vs receptors** (STBD1, NBR1, LMNB1): absence of direct cargo-receptor complexes would corroborate "no selectivity role."

Curation Leads (require curator verification)

- **Candidate action:** Keep GO:0000425 / GO:0044804 / GO:0061723 as **IBA/non-core, shared-machinery** annotations; add an annotation note that WIPI2 acts as generic autophagosome-biogenesis machinery, not a cargo receptor. Do **not** upgrade to experimental; do **not** remove outright (each IBA has a valid within-family experimental seed).
- **Candidate core terms to foreground instead:** MF GO:0032266 (PI3P binding), MF GO:0030674 (protein-macromolecule adaptor activity), BP GO:0000045 (autophagosome assembly), BP GO:0016236 (macroautophagy), CC GO:0000407 (phagophore assembly site) — all direct-evidence supported. Prefer these over "protein binding".
- **Candidate references with exact snippets to verify:**

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— "Atg16L1 directly binds WIPI2b ... WIPI2b is a PtdIns(3)P effector upstream of Atg16L1

and is required for LC3 conjugation ... WIPI2b is also required for autophagic clearance of pathogenic bacteria."

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 — "LC3 lipidation is strictly dependent on the recruitment of WIPI2 that in turn depends on PI(3)P ... WIPI2 allosterically activate[s] the E3 complex."
- P 24265594

 — Drosophila core Atg genes (incl. Atg18a) required for autophagic glycogen breakdown (glycophagy IBA seed).
- P 26524528

 — "The autophagy protein LC3/Atg8 ... directly interacts with the nuclear lamina protein lamin B1" (nucleophagy selectivity is LC3–cargo, not WIPI2).
- P 39236246

 — STBD1 CBM20/LIR mechanism of glycophagy selectivity.
- **Suggested question for curator:** Does this review's policy annotate core autophagy machinery to each selective-autophagy sub-process (IBA), or restrict such terms to factors with pathway-specific experimental evidence? The answer determines retain-as-non-core vs generalize.
- **Suggested experiments:** the four discriminating tests above.

Provenance

Machine-readable artifacts: `artifacts/WIPI2_evidence_matrix.csv`, `artifacts/WIPI2_GO_decision_table.csv`.

Computed with QuickGO REST (annotation + ontology endpoints) and PANTHER geneinfo, executed in-session (iterations 1–3): - WIPI2 annotation pull (50 rows) → three target terms are IBA only. - GO term definition pull → pexophagy/glycophagy "by macroautophagy"; nucleophagy broadly defined (not yeast-PMN-restricted). - Paralog scan (WIPI1/2/3/4 + yeast Atg18/Atg21/Hsv2) → identical blanket IBA; yeast Atg18 IMP seeds pexophagy/nucleophagy. - Cross-family experimental-source scan → glycophagy seeded by Drosophila Atg18a

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human experimental annotations for all three terms are on dedicated receptors/regulators, never WIP12.

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